

Lab 3: Architectural Justification

Municipal Solid Waste Route Optimizer (Problem Statement #25)

Architecture Selection: We chose Microservices Architecture for the Municipal Solid Waste Route Optimizer System.

Architectural Choice

Layered, Client-Server and Microservices styles were compared. The system is decomposed into six components behind an API Gateway: Web Portal UI, API Gateway, Bin Telemetry Service, Route Optimizer Service, Manifest Service and Complaint Service. Smart bin sensors are external field devices. Components collaborate only through the interfaces IPortalAPI, IRoutePlanning, IRouteData, IManifest, IComplaint, IBinStatus and ITelemetryIngest, and each service owns its own datastore.

Two Reasons for the Choice

- **Workloads differ sharply and must scale independently.** Telemetry ingestion is a constant stream from up to 10,000 bins, route recalculation is a CPU-heavy burst (NFR: 50 trucks, 10,000 bins in under 45 s), while complaint logging is light and citizen-driven. Separate services let us scale the Route Optimizer and Telemetry Service without over-provisioning the rest. A layered monolith would force scaling everything together.
- **Fault isolation keeps critical operations running.** If the optimizer is overloaded or being redeployed with a new algorithm, drivers can still read their existing manifests and citizens can still file missed-pickup complaints. A client-server design with one central server would make any failure a system-wide outage. Loose coupling via interfaces also means a service (e.g., a new sensor vendor's ingestion) can be replaced without touching the others.

Security Advantage

All Sanitation Supervisor and Truck Driver requests enter through the single API Gateway, which performs authentication (JWT) and role-based authorization, so a driver can only reach manifests while a supervisor can trigger route generation. Internal services are not exposed publicly, sensor traffic is encrypted (MQTT over TLS), and a compromised service cannot read another service's data because each has a private datastore, limiting the blast radius.

Performance Benefit

Horizontal scaling of the Route Optimizer Service (multiple instances, each handling a subset of the fleet or zones) helps meet the 45-second recalculation target under peak load. Telemetry ingestion is decoupled from route computation, so a burst of sensor readings never blocks the optimizer or portal, and the optimizer reads pre-filtered bin status (bins at or above 80% fill) through the IBinStatus interface.

Trade-off Acknowledged

Microservices add operational complexity, network latency between services and data-consistency challenges. For a city-wide fleet system these costs are justified by the scaling and availability needs above, and are mitigated by using gRPC for internal calls and a single gateway entry point.